III. Suggestions of Correlation:

- What is correlation? What do you understand by partial correlation and multiple correlations? (RU: 2011, -ICE)
- Define positive and negative correlation with example. (RU: 2014, -ICE)
- What is simple correlation coefficient? What does it measure? Discuss the situation, when r = 1, -1 and 0. (RU: 2011, -ICE)
- What is a scatter diagram? What is the important of scatter diagram? (RU: 2014,11 -ICE)
- Explain different types of correlation with examples. Mention important properties of correlation coefficient.
- Define coefficient of correlation.
  Show that the correlation coefficient lies between –1 and +1. (RU: 2014, 2011, 2012-ICE)
- Show that correlation coefficient is independent of change of origin and scale of measurement. Explain the value of r (correlation coefficient) when it takes -1, 0, 0.2, 1
  (RU: 2015,2013, -ICE) CSE :2017,2015,2013,2011

- **Karl Pearson Correlation: MATH**

(01). Calculate coefficient of Correlation between price and supply. Interpret the value of correlation coefficient.

Price	8	10	15	17	20	22	24	25
Supply	25	30	32	35	37	40	42	45

(02) Find the correlation coefficient between the marks obtained by ten students in economics and statistics. (RU: 2014, -ICE)

Marks (in economics)	70	68	67	55	60	60	75	63	60	72
Marks (in statistics)	65	65	80	60	68	58	75	62	60	70

(03). Define correlation and regression. Find out correlation coefficient between height of father and son from the following data. (RUET: 2009,2011,2012-civil) (RU: 2013, -ICE)

Height of father	65	66	67	68	69	70	71	67
Height of son	67	68	64	72	70	67	70	68

(04). Calculate correlation coefficient between the heights (in cm) and weights (in kg) of 10 students given below . (RU: 2013, -ICE)

Heights (in cm)	150	155	157	160	162	165	168	172	175	180
Weights (in kg)	55	58	60	61	65	62	74	75	80	89

RANK CORRELATION:

- 9. What is Rank correlation? How does it differ from ordinary correlation? (RU: 2015, -ICE)

- 10 Discuss the concept of Rank correlation? Show that spearman's rank correlation coefficient is given by (RU: 2013, -ICE, CSE : 2016)

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d^2}{n(n^2-1)}$$

(Usual notation are used)

(5). Two judges gave the following ranks to eight competitors in a beauty contest. Examine the relationship between their judgments.

Judge A	4	5	1	2	3	6	7	8
Judge B	8	6	2	3	1	4	5	7

(6). From the following data, calculate the coefficient of rank correlation.

X	36	56	20	65	42	33	44	50	15	60
Y	50	35	70	25	58	75	60	45	80	38

(7). Find the Rank correlation from the following data. (RUET: 2015-civil)

X	16	19	25	28	25	30	32	36
Y	192	218	210	232	218	236	210	236

(8). Two judge A and B in a beauty contest rank the ten competitors in the following order.. . (RUET: 2014-civil)

A	4	3	1	2	7	9	8	10	5	6
B	1	6	7	5	8	10	9	3	2	4

Do the judge appear to agree in their standard.

(9). Find the rank correlation coefficients of the marks in two test.

Marks in Test I	70	68	67	65	60	60	75	63	60	72
Marks in Test II	65	65	80	60	68	58	75	62	60	70

iv. Suggestions of Regression:

- Define regression. What are the uses of regression analysis? Describe different types of regression.
- Discuss the properties of regression coefficient. (RU: 2014--ICE)
- "If one of the regression coefficient is greater than unity , the other must be less than unity " '---Justify. RU: 2014--ICE)
- What is regression? How does it differ from correlation?
- Consider a regression model of Y on X derive the least square estimates of the model parameter. (RU: 2013--ICE)
- Consider the Simple linear regression model

  $Y_i = \alpha+\beta x_i+\varepsilon_i$: i=1,2,3,……,n

  Stating the necessity assumption. Find the least square estimates of α and β. (RU: 2014--ICE)
- What do you mean by regression model and regression line .Distinguish between correlation and regression? (RU: 2015, -ICE)
- What do you mean by regression line of Y on X and regression line of X on Y? From the following data, find the regression equation

  ΣX = 21, ΣY = 20, ΣX2 = 91, Σ XY = 74, n = 7

- **Regression : MATH**

(1). From the following data find the least square regression line of X and Y. (RUET: 2011-civil)

X	65	66	67	68	69	70	71	67
Y	67	68	64	72	70	67	70	68

(2). Estimate the regression equation of Y on X for the given data (RU: 2013--ICE)

Family members (X)	4	7	3	8	5	4	6
Expenditure on food in taka(Y)	160	80	75	90	140	80	150

(3). Consider the following data set on two variables X and Y:

X	2	3	4	5	6	8
Y	7	6	5	5	3	2

(i) Find the equation of regression line of Y on X and comment.
(ii) Estimate the value of Y when x=10. (RU: 2015, -ICE)

(4). To study the tensile strength of a certain type of wore the following pairs of observations were recorded. Where X is the diameter in cm and Y is the mass supported in kg' cm

X	0.7	0.8	1.0	1.2	1.4	1.6	1.8	2.0	2.2	2.3
Y	14	26	35	56	42	98	82	88	134	124

(i) Find the equation of regression line of Y on X for the above data .

(ii) Estimate the value of Y when x=2.5.and x=2.7

(RU: 2012, 2011--ICE)